Project Management Plan

for

THEIA

Prepared by Jackson Pickett, Faisal Kherwish, Hank Corrion

Pancake Patrollers

10/18/2026

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Revision History

Name	Date	Reason for Changes	Version
Jackson Pickett, Faisal Kherwish, Hank Corrion	9/20/2026	initial draft	1.0
Jackson Pickett, Faisal Kherwish, Hank Corrion	9/20/2026	revised	2.0
Jackson Pickett, Faisal Kherwish, Hank Corrion	10/18/2026	Final revision	3.0

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1. Overview

Project THEIA is a smartphone app intended to help visually impaired people with navigating indoor environments. Indoor environments include university buildings, classrooms, offices, bathrooms, lounges, elevators, multiple floors, and connected buildings. The app will provide navigation assistance for users who can't rely on a visual interface.

Phase I of the project focuses on software requirements engineering. The team will analyze the requirements for THEIA, identify unresolved issues, develop improved and traceable requirements, construct AS-IS and TO-BE scenarios, develop a prototype and user manual, and document the team's project management.

1.1. Project Purpose, Objectives, and Success Criteria

The app's primary use is to assist a user in navigating from one indoor location to another, potentially including locations on different floors or in connected buildings. The system may determine routes, provide navigation instructions, detect obstacles, assist with turns, and provide emergency functionality. Some important non-functional objectives include safety, navigation speed, comfort, usability, and customization.

1.2. Project Deliverables

Deliverable	Delivery Date	Delivery Method	Comments
AS-IS/TO-BE Scenarios and Presentation	10/18/2026	Canvas/Git Repository	Presentation slides and recorded presentation
WRS Document - Issues	10/18/2026	Canvas/Git Repository	Identify incompleteness, inconsistency, ambiguity, redundancy, unsoundness, and other issues
WRS Document - Improved Understanding	10/18/2026	Canvas/Git Repository	Problems, goals, objectives, requirement specifications, and traceability
THEIA Prototype	10/18/2026	Canvas/Git Repository	Phase I mock up or GUI demonstration
Preliminary User Manual	10/18/2026	Canvas/Git Repository	Instructions
Revised Project Plan	10/18/2026	Canvas/Git Repository	Team organization, task distribution, responsibilities, and Phase I schedule

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AI Usage and Teamwork Report	10/18/2026	Canvas/Git Repository	Documentation of AI use and teamwork

1.3. Assumptions, Dependencies, and Constraints

AS-1: Team members will have regular access to the course materials, project templates, and THEIA Preliminary Definition

AS-2: Team members will communicate regularly and contribute to the development and review of the project deliverables

AS-3: The Phase I prototype is expected to demonstrate the team's vision for THEIA rather than a complete app.

DE-1: Development of the WRS document depends on analysis of the THEIA Preliminary Definition.

DE-2: The improved requirements depend on the issues and resolutions identified during requirements analysis

DE-3: The prototype and user manual depend on the TO-BE scenarios and requirements

DE-4: Completion of the final deliverables depends on coordination and review between team members

CO-1: All Phase I work must be completed according to the deadlines in the course schedule

CO-2: The project focuses on requirements engineering instead of implementation of a complete app

CO-3: The team must maintain project materials in the Git repository as required by the assignment

CO-4: AI generated material must be reviewed by team members before being incorporated into project deliverables

1.4. References

No References at the moment but might be added in the future after a revisions

1.5. Definitions and Acronyms

THEIA - Proposed smartphone app for helping visually impaired users with indoor navigation

WRS (World, Requirements, and Specification - the requirements document used to establish relationships and traceability between the problem domain and system requirements

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AS-IS Scenario - Description of how a stakeholder performs an activity or encounters a problem without the system

TO-BE Scenario - Description of how the activity could occur after introducing THEIA

FR (Functional Requirement) - describes functionality or behavior expected from the system

NFR (Non Functional Requirement) - describes system qualities or constraints such as safety, usability, or performance

Traceability - The ability to establish relationships between problems, goals, objectives, and requirement specifications

Stakeholder - A person or organization affected by or interested in the THEIA system

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2. Project Organization

The THEIA project will be completed by the team. Although individual members will have primary responsibilities for specific deliverables and tasks, major decisions and final deliverables will be reviewed collectively to maintain consistency across the project.

2.1. Process Model

The team will be using an iterative requirements engineering process during Phase I. The team will begin by looking at the THEIA Preliminary Definition and identifying issues with the initial requirements. Proposed resolutions will then be discussed and then incorporated into an improved understanding of the system. The general process will look something like:

Preliminary Definition → Requirements Analysis → Issue Identification → Issue Resolution → Improved Requirements → Traceability → Scenarios and Prototype → Review

2.2. Organizational Structure

The team will be using a collaborative structure with a team liaison serving as the primary point of contact between the team and instructor. Individual members will serve as primary owners of assigned tasks and deliverables, while all members will participate in review, communication, and integration of final materials

2.3. Roles and Responsibilities

Team Member	Role	Primary Responsibilities
Jackson Pickett	Team Liaison	Instructor communication, team coordination, scheduling tracking
Jackson Pickett	Project Manager	Project tracking and assisting in other tasks
Faisal Kherwish	Requirements/Issues Lead	Preliminary definition analysis, WRS issues and resolutions
Faisal Kherwish	Requirements/Traceability Lead	Improved understanding, problems/goals/objectives/rs, traceability
Hank Corrion	Scenario/Presentation Lead	AS-IS/TO-BE scenarios and presentation
Hank Corrion	Prototype/User Manual Lead	Prototype development and preliminary user manual
All Members	Reviewers	Review requirements, attend meetings, resolve inconsistencies, review final submissions
All Members	Documentation	Revised Project Plan, AI use, and teamwork report

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3. Managerial Process Plans

3.1. Management objectives and priorities

The management philosophy for this project is to break the entire project into smaller tasks with achievable deliverables. Tasks will be divided up and assigned through communication with the team via Discord or email.

The deliverables for Phase 1 are due mid October so the team will make sure to complete the drafts before the deadline so time for review is possible. Keeping all the deliverables accurate and consistent is a top priority.

Members effectively communicating their progress on their tasks is to be prioritized as well. Team meetings will be a time for members to come forward with their problems and status of their designated tasks. The team liaison will be in charge of any and all communications with the instructor and passing that feedback back to us.

Using AI responsibly is a must. AI generated material must be reviewed by team member(s) before it is finalized to be used in the project.

3.2. Assumptions, dependencies, and constraints

Each team member will probably have a time constraint given that they all have full schedules. Weekly meetings with team members to keep our progress evaluated is a good solution to make sure everyone is on track.

3.3. Risk management

No.	**Risk**	**Type**	**Likelihood**	**Description**
1	Failure to meet deadlines for deliverable	Managerial	Low	Failing to produce a satisfactory deliverable on time.
2	Under viewing AI generated material	Technical	Medium	Lack of reviewing material generated by AI can lead to issues in the app development process.
3	Requirements Change	Technical	Low	Unless stated by the instructor, requirements for the project itself shouldn’t change.
4	Lack of team member communicati	Managerial	Likely	Team members might lack communication between each other due to personal reasons.

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No	Risk	Monitoring and Controlling
1	Failure to meet deadlines for deliverable	● Keep communication crystal clear when it comes to status on deliverables prior to the deadline
2	Under viewing AI generated material	● Precise review of everything resulting of an AI prompt ● Creation of very precise prompts that will not generate malicious content
3	Requirements Change	● Constantly check email and canvas for any updates on the project requirements and specifications ● Assign more resources to meet the additional criteria
4	Lack of team member communication	● Weekly meetings between members ● Consistent talks and questions in our group chats or emails

4. Technical Process Plans

Phase 1 will be focused on requirements engineering and documentation rather than building the full application. This section covers the tools and methods that the team will use and documents we aim to produce.

4.1. Methods, tools, and techniques

Development will be hardware and OS agnostic and each member will use their own development environment. Documents will be written in Google Docs and stored in the team's Git repo. The prototype will be made in Figma for mobile screens and the presentation will be made in Google Slides and recorded on Zoom. The team will communicate through Discord and email. We will follow the iterative process described earlier in section 2.1 and label each requirement as functional or non-functional. Every deliverable will be reviewed by at least one member who did not write it and requirements will be checked against the requirements review checklist. Any AI generated material will be reviewed by a team member before it is used.

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The planned tech stack is React Native with TypeScript for the app and Firebase for the backend. The app will run on iOS and Android phones and will need an internet connection. Code will be written in VS Code and tested on phones and emulators. The app will not be built during Phase 1. Scenarios and system behavior will be modeled with UML use case and activity diagrams made in draw.io. Documents will follow the templates and all code will follow a style guide. New work will be merged through reviewed pull requests and review comments and bugs will be tracked as issues in the repo. Unit, integration, and system testing are planned for later phases, with automated builds and tests running on each pull request.

4.2. Software documentation

Document	Template or Standard	Created By	Reviewed By	Target Date	Distribution
AS-IS/TO-BE Scenarios and Presentation	Google Slides	Hank Corrion	Faisal Kherwish, Jackson Pickett	10/16	Canvas, Git Repo
WRS Document - Issues	Requirements review checklist	Faisal Kherwish	Jackson Pickett, Hank Corrion	10/16	Canvas, Git Repo
WRS Document - Improved Understanding	WRS model	Faisal Kherwish	Jackson Pickett, Hank Corrion	10/16	Canvas, Git Repo
THEIA Prototype	Figma	Hank Corrion	Jackson Pickett	10/16	Canvas, Git Repo
Preliminary User Manual	Canvas template	Hank Corrion	Jackson Pickett	10/16	Canvas, Git Repo
Revised Project Plan	Canvas template	Jackson Pickett	Faisal Kherwish, Hank Corrion	10/16	Canvas, Git Repo
AI Usage and Teamwork Report	Canvas template	All members	Jackson Pickett	10/16	Canvas, Git Repo

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